ENVISION MISSION TO VENUS: SUBSURFACE RADAR SOUNDING

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ABSTRACT

This paper presents the Subsurface Radar Sounder (SRS) instrument onboard European Space Agency's (ESA) EnVision mission. EnVision is one of the three candidates selected for the Cosmic Vision 2015-2025 M5 medium-class missions. It is aimed at exploring the activity, the geologic history and the atmosphere of Venus. SRS is an orbital ground-penetrating radar with the unique science objectives of understanding the evolution of Venus' surface by searching for subsurface dielectric interfaces in the top hundreds of metres of the crust. In the paper, we describe the main science objectives of SRS, the performance evaluation under expected target conditions, the instrument design and the acquisition strategy that maximize the scientific returns.

Index Terms— Radar sounder, Venus, subsurface geology, planetary exploration

1. INTRODUCTION

Venus, our closest neighbor in the solar system and the most Earth-like planet, evolved to be an unusual terrestrial planet having unique surface morphologies and a thick atmosphere. Exploration of Venus can reveal evidences of the current geological activity, the processes that transformed the planet's surface, and the hostile atmosphere. With the aim of exploring Venus, EnVision is selected as one of the three candidates competing for ESA's Cosmic Vision 2015-2025 M5 medium-class down-selection, with a scheduled launch in 2032.

The scientific goals of the mission are focused on addressing three key questions – (1) How geologically active is Venus today? (2) How did Venus' surface and interior evolve? and (3) Why did Venus' climate become so hostile? To support the achievement of the aforementioned objectives, EnVision will carry on-board three spectrometers (VenSpec-U, VenSpec-H and VenSpec-M) for mapping the constituents of the atmosphere and the surface minerology; a synthetic aperture radar (VenSAR) for imaging the surface morphology at high resolution; and a subsurface radar sounder (SRS) for direct measurements of the subsurface. Moreover, a radio science and gravity experiment (RadioScience) is planned.

This paper focuses on the SRS for subsurface observations, which address the main objective of understanding the geological history of Venus. The objectives of SRS are to study the evolution of the Venusian crust by searching for evidence of buried features beneath the surface. SRS is a nadir-looking radar sounder instrument that operates at 9 - 30 MHz frequencies. The signal attenuation at these frequencies for dry terrestrial material (silicates) is low enough to allow a penetration of a few hundred metres (up to a kilometre, depending on the central frequency and the target material) into the subsurface. As it propagates through the subsurface, the signal is reflected back at interfaces of significant dielectric contrast, which appear as peaks in the received signal. The profile of the target along the radar track is processed to obtain a depth image of the subsurface, known as the radargram, which shows buried structures and compositional variations beneath the surface.

Several radar sounders have revolutionized the scientific understanding of the processes occurring on different planetary bodies. On Mars, we have the Mars Advanced Radar for Subsurface and Ionospheric Sounding (MARSIS) and the Shallow Radar (SHARAD). The Moon has been probed by the Lunar Radar Sounder (LRS). These sounders have resolved the ambiguities in the geological interpretation of the Moon and Mars by providing direct measurements of the subsurface and producing ground-breaking scientific results. For the study of the Jovian icy moons, currently two radar sounders are under development, the Radar for Icy Moons Exploration (RIME) [1] and the Radar for Europa Assessment and Sounding: Ocean to Near-surface (REASON).

SRS will follow in the heritage of these previous radar sounders and will be the first instrument to profile the subsurface of Venus. SRS can acquire fundamental information on subsurface geology by mapping the vertical structure (mechanical and dielectric interfaces) and properties of the tesserae, buried craters plains, lava flows and volcanic features, thus providing useful data for inferring their genesis. It can also provide information on the surface in terms of roughness, composition and dielectric properties at wavelengths completely different from those of SAR, thus allow-

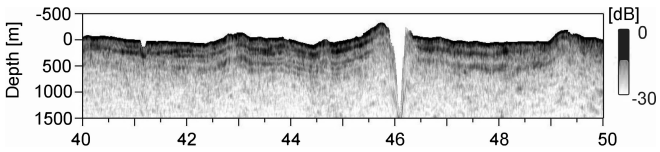
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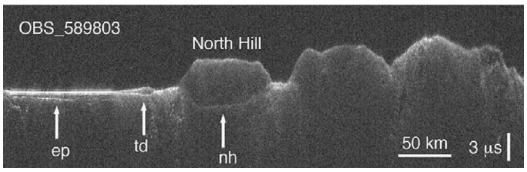
ing a better understanding of the surface properties. Furthermore, SRS will provide surface altimetry measurements to model the surface topographic variations. SRS will achieve its goals by acquiring: (i) low density lossy compressed data with a global coverage of the planet with the goal of searching for possible subsurface dielectric interfaces, and (ii) high density and non-compressed data with regional coverage over high-interest targets.

2. SCIENCE GOALS

Table 1 shows the main scientific objectives of the EnVision mission that will be supported by SRS measurements and are related to planetary geology. Geological features similar to the Venus subsurface targets have been detected by radar sounders on other planetary bodies, and thus indicate promising returns from SRS profiles. Some of the interesting examples presented in Table 1 are shown in Figure 1. SRS will search for stratigraphic horizons, possibly representing different episodes of the volcanic history of Venus (e.g. Fig. 1(a)). There are several debates regarding the style of volcanism on Venus, one of them being the explosive eruptions that result in the formation of highly porous pyroclastic flow deposits, which can bury the pre-existing terrain. SRS would search for the base of such pyroclastic deposits (e.g. Fig. 1(b)).



(a)



(b)

Fig. 1: Examples of detection of subsurface geological features similar to some of the Venus targets by radar sounders on other planetary bodies: (a) stratigraphy of lunar mare detected by LRS [2], (b) base of pyroclastic deposits indicated by the arrows, as detected by SHARAD [3].

3. DESIGN OF SRS PARAMETERS

In order to meet the science goals presented in Section 2, SRS is being designed as a nadir-looking radar sounder instrument. The current baseline configuration considers a central frequency of 9 MHz and a bandwidth of 6 MHz, for which the estimated antenna dipole size is 16 m. Studies are under development for possible tuning the parameters in the range 9-30 MHz. The final design of the SRS instrument will be a compromise to account for environmental (e.g, cosmic noise, ionosphere), target (e.g, surface and subsurface scattering properties), the orbit characteristics (i.e. altitude, spacecraft velocity, orbit eccentricity), and resources constraints (e.g, mass and power).

The SRS scientific requirement of having a vertical resolution in the order of 20 m and a penetration depth in the order of hundreds of metres results in the design of a radar sounder system with a large fractional bandwidth. The selection of a lower central frequency, within the specified range, is likely to increase the penetration depth of the instrument with respect to central frequencies in the higher part of the considered range. However, a higher central frequency allows to increase the resolution of the instrument and reduce the dipole antenna size that also depends on the central frequency. Table 2 summarizes the main SRS parameters.

With respect to previous deployed radar sounders, the high fractional bandwidth represents a surmountable technical challenge for the instrument design in terms of achieving an optimal matching across the entire radar bandwidth. The design of SRS will allow a flexible configuration of some radar parameters like pulse repetition frequency, pulse length, receiver window length, quantization levels and lossy compression factor even during the mission operation. Nevertheless, to simplify the resource allocation planning and the acquisition strategy, two fixed different acquisition modes are considered obtaining a datarate in the range reported in Table 2.

4. PERFORMANCE EVALUATION

The strategy for designing the SRS instrument is based on the performance evaluation for the expected conditions during acquisitions on Venus. The performance of SRS can be quantified in terms of penetration depth, and range, along-track and across-track resolutions. These can be described as a function of the target dielectric properties, the background noise, the ionospheric effects, the surface topography, and the instrument parameters.

The interpretation of Venus' surface from past missions presents a wide range of possible geological composition and thus, of dielectric properties of the near-surface. Due to the presence of many volcanic features, the surface is likely to be dominated by volcanic rocks, such as basalts (real permittivity in the range 4-10), in the form of a matrix loaded with small fragments of conducting material. Some studies argue that the real permittivity of the surface is likely to be similar to the Moon, although the regolith layer could be thinner and having a higher loss tangent. The attenuation is dependent on the high surface temperature ($\sim$740 K) and the radar frequency. Some regions are interpreted to have a morphology similar to

Table 1: Main SRS scientific objectives and examples of relevant similar objectives and related targets detected by other radar sounders

Mission goal	**SRS objectives**	**Examples from other sounders**
Study the impact craters and their modification, to establish absolute age of geological units	Buried impact craters, Layering in modified craters	MARSIS buried basins LRS mare filled crater [2] LRS sequences in Smythii basin
Establish stratigraphic relationships between different geologic units	Buried strata embayed older terrain, Stratified or layered units	SHARAD buried volcanic vents LRS stratigraphy in lunar mare [2]
Understand the origin of the tessera highlands	Buried structures in tesserae highlands	LRS older S11 reflector extending below younger unit
Understand the nature of the plains	Buried structures in plains, Cross-cutting relationships	
Characterize volcanic features	Base of lava flows, Base of pyroclastic deposits, Buried lava tubes, Buried lava chambers	SHARAD lava flows in Ascraeus Mons SHARAD pyroclastic deposits in MFF [3]
Characterize the styles of tectonic deformations on Venus	Buried features	SHARAD interface beneath wrinkle ridges

Table 2: Current parameters of the proposed SRS instrument.

Instrument Parameter	**Parameter Value**
Central frequency	In the range 9 - 30 MHz (Baseline 9 MHz)
Bandwidth	In the range 6 - 20 MHz (Baseline 6 MHz)
Antenna type	Dipole (deployable)
Antenna size	$\leq$ 16 m (depending on central frequency)
Peak radiated power	$\leq$ 200 W
Vertical resolution in medium	16 m ($B = 6$ MHz) – 5 m ($B =$ 20 MHz) (for $\mathrm{Re}\{\epsilon_r\} = 6$)
Data rate	In the range 3.8 - 7.6 Mbit/s

pyroclastic flow deposits, which are highly porous and thus, likely to be highly radar transparent in the HF bands.

For the performance evaluation, we have categorized the different hypotheses on the Venusian target dielectric properties, ranging from highly porous low-loss deposits ($\mathrm{Re}\{\epsilon_r\} \sim 2, \tan\delta \sim 0.01$) to highly conductive lossy volcanics ($\mathrm{Re}\{\epsilon_r\} \sim 10, \tan\delta \sim 0.1$). We also considered different dielectric contrast values at the subsurface interface of interest, which depend on the nature of emplacement. For example, on the one hand, the dielectric contrast between two overlapping lava flows representing eruptions having very different compositions (such as the fraction of Fe, Ti oxides) could be very high. On the other hand, the weathered soil formed over its parent material would represent a low contrast interface. Thus, a wide range of contrast between 0.2 and 2 in the real permittivity is considered for the performance analysis.

Radar sounding of Venus at MHz frequencies will be affected by three main sources of noise: (1) the receiver thermal noise, (2) the isotropic galactic noise, and (3) the bursts of solar radio emissions. For a central frequency of 9 MHz and a bandwidth of 6 MHz, the galactic noise is evaluated to be 105 dB. Out of the three sources of noise, the solar radio emissions are expected to dominate the SNR in the conditions of high solar activity. Studies are in progress to characterize the solar emissions expected at the time of the EnVision operation. However, the effects can be mitigated with night time acquisitions.

Radio propagation through the ionosphere can distort the received signal by introducing echoes' time delays and causing the loss of peak power and range resolution. A preliminary assessment has been performed to determine the value of the Venus ionospheric cut-off frequency. Different electron density profiles have been considered where both low and high solar fluxes are present [4]. The cut-off frequency is estimated to be around 1 MHz for the night side and and in the range between 2 and 7 MHz on the day side depending on the solar zenith angle.

Data from the Pioneer Venus Orbiter [5] and the Magellan spacecraft suggest that most of Venus is characterised by RMS slopes θ in the $[1^\circ, 10^\circ]$ range for baselines of a few tens of centimetres to a few tens of metres. Assuming 10 m (as worse case), we obtain the RMS height difference as $\sigma = 10\tan(\theta)$ [6], leading to the range $[0.17\text{ m}, 1.7\text{ m}]$. With respect to the power P_0 backscattered from a perfectly smooth surface, the power P backscattered from a rough surface will be modelled as $P = \chi^2 P_0$. Assuming (simplisti-

cally) that the terrain can be modelled as a stationary Gaussian random rough terrain, the coherent losses can be estimated as $\chi^2 = \exp[-(4\pi\sigma/\lambda)^2]$ under the small perturbation model (SPM). This leads to coherent losses between 0.6 and 5.7 dB.

Table 3 shows the estimated range of performances at the baseline instrument (central frequency of 9 MHz and bandwidth of 6 MHz) computed for the range of considered target properties and the background noise power. The penetration depth is estimated assuming an SNR margin of 5 dB to account for the scattering and ionospheric losses. The range resolution in free space is evaluated for a 6 MHz bandwidth, assuming that the sidelobe weighting function results in at most 67% enlargement of the main-lobe. Note that the interesting target subsurface interfaces described in Table 1 that are relevant for the science objectives of SRS, in average, occur at a depth of a few hundred metres. The penetration and range resolution performances show that SRS is able to detect these interfaces with an appropriate SNR.

Table 3: Range of variability of the expected SRS performances for the baseline configuration considering all the range of target and noise component values.

Performance parameter	**Range of values**
Maximum penetration depth	100 - 1300 m
Range resolution in free-space	42 m
Range resolution in medium	15 - 30 m
Along-track resolution	4 km
Across-track resolution	7 km

5. ACQUISITION STRATEGY

The scientific goals of the mission require subsurface sounding over the entire surface of the planet in order to search for buried features and understand in three dimensions the boundaries between units. Moreover, more detailed acquisitions are required on specific areas known to be of high interest due to their geological characteristics. Given these requirements, the acquisition strategy of SRS is composed by two acquisition modes: high density and low density.

The high density mode is intended to be used to cover selected targets with expected subsurface interfaces by acquiring raw data on tracks spaced by 10 km over a total of 20% of the surface. The residual 80% of the planet will be acquired in low density mode. This mode leverages on an acquisition track spacing of 50 km and a lossy compression based on pre-summing to search, at global scale, for possible targets with subsurface interfaces, limiting the overall generated data-volume.

The favorable conditions for SRS acquisitions depend on the ionosphere and the solar activity. We analyzed them by considering the solar zenith angles during the possible orbits, given a minimum mission length of 4 Venus sidereal days (2.7 Earth years). To maximize the detection performances, the acquisitions will be preferred on the night side. Due to the mission length, a fraction of the surface is only observable while on the dayside. However, our analysis shows that, the solar zenith angle of these acquisitions is always greater than 40° (more than 80° for 45% of the dayside acquisition), thus limiting the ionospheric impact.

6. CONCLUSIONS

The SRS measurements of subsurface profiles and surface altimetry of Venus have the potential to address many crucial science objectives related to the geological past of Venus, which can fundamentally improve our understanding of the processes occurring in the Solar System and provide ground breaking science returns. This paper has presented the science objectives, the performance analysis, the range of design parameters and the acquisition strategy of SRS that drive its development. Further studies are in progress to better constrain the expected acquisition conditions, in particular the complex dielectric permittivities at high temperatures, the nature of the solar radio emissions and the ionospheric characteristics. This will guide the final tuning of the SRS instrument parameters and design.

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